

Full Analysis Document

AI-Powered Dynamic Mental Health Monitoring and Distress Prediction System for Victims of Atrocities

Ministry of Social Justice and Empowerment (MoSJE) • Internal hackathon: 29 Aug 2026

Contents

- 1. Executive Summary
- 2. Who This Is For
- 3. Glossary & Domain Context
- 4. Stakeholder Map
- 5. Scope & Intake Channels
- 6. End-to-End Scenario Walkthrough
- 7. Mandatory Requirements (SIH's Expected Solution)
- 8. Feature Roadmap
- 9. Feature Parity — Mobile App & Web App
- 10. Competitive Landscape — What Already Exists
- 11. Integration & Value-Add Strategy
- 12. Database Design (Scoped to Mandatory Features)
- 13. AI/ML Model Architecture
- 14. Training Data Strategy
- 15. Legal, Privacy & Compliance
- 16. System Safety Design
- 17. Assumptions & Open Questions for Team Discussion
- 18. Sources

1. Executive Summary

This PS asks for an AI layer sitting on top of India's existing atrocity-victim support infrastructure (NHAA 14566 and related channels) that continuously monitors victim psychological well-being, predicts crisis escalation before it happens, and routes timely interventions through counsellors and officials — while staying explainable, private, and secure.

No existing product does this end to end (Section 7). The pieces exist separately, but nothing stitches helpline-style intake, trauma-specific distress prediction, and authority escalation together for this population — that gap is the core of what this PS is asking your team to build.



2. Who This Is For

Category	Who	Details
Primary beneficiary (monitored)	SC/ST atrocity victims & complainants	Priority cases: victims of rape/gang rape, murder, grievous hurt, arson; intimidated witnesses; families affected by caste-based violence

Category	Who	Details
System operators	Counsellors, district/state welfare officials	Receive alerts, action interventions, work the case queue
Policymakers	State/national administrators	Consume aggregate dashboards for planning and oversight
Coordination	Law-enforcement agencies	Coordinated response on escalated cases
Commissioning body	Dept. of Social Justice and Empowerment, MoSJE	Government infrastructure, not a consumer product

3. Glossary & Domain Context

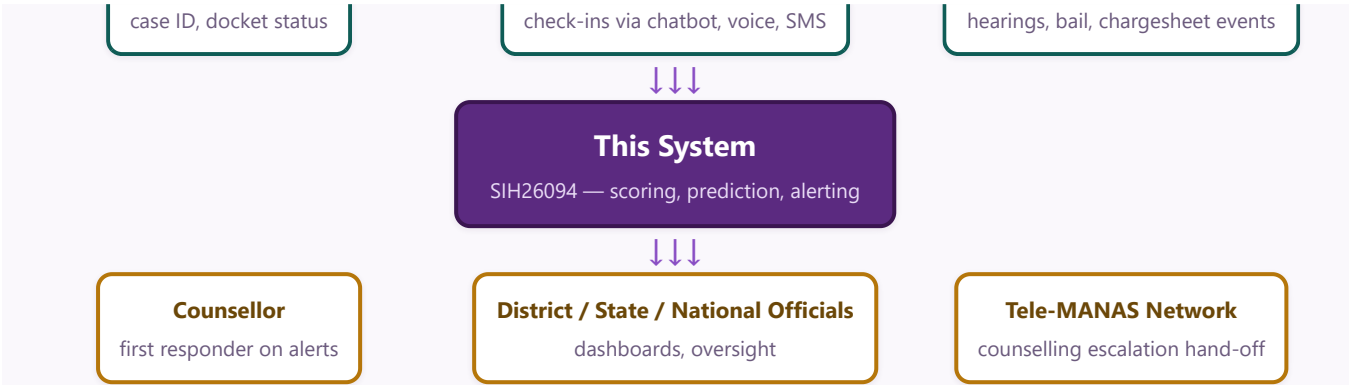
Term	Meaning
SC/ST (PoA) Act, 1989	The Scheduled Castes and Scheduled Tribes (Prevention of Atrocities) Act — the central law defining what counts as a caste-based "atrocitiy" and mandating relief, compensation, and time-bound investigation/trial
NHAA	National Helpline Against Atrocities (14566) — MoSJE's helpline for SC/ST atrocity victims
SAMBAL	The web portal paired with NHAA, tracking FIR/investigation/chargesheet status
Tele-MANAS	Ministry of Health's separate national tele-mental-health helpline (14416) — general-purpose, not atrocity-specific
Docket Number	The unique ID assigned to a registered complaint, tracked through FIR → investigation → chargesheet → trial → compensation
IVRS	Interactive Voice Response System — automated phone menu/call system, no smartphone required
ASR / TTS	Automatic Speech Recognition (voice-to-text) / Text-to-Speech (text-to-voice)
NLP	Natural Language Processing — the AI field concerned with understanding human language
XAI	Explainable AI — techniques that make an AI decision interpretable to a human, instead of a black box
DPDP Act, 2023	India's Digital Personal Data Protection Act — governs consent, data minimization, breach notification
Dynamic Distress Score	This PS's core metric — psychological distress tracked as a trend over time, not a single test
PII	Personally Identifiable Information — data that can identify a specific person

4. Stakeholder Map

NHAA / SAMBAL

Victim / Complainant

Judiciary / Case Records



5. Scope & Intake Channels

Channel	Type	Notes
NHAA Helpline (14566)	Voice	Toll-free, round the clock, multilingual
Integrated Portal	Web	Self-service complaint/status portal
Chatbot	Text / Voice	Primary AI-driven check-in surface
Mobile application	App	Primary victim-facing touchpoint
IVRS	Automated voice	For non-smartphone users
SMS	Text	Follow-up / low-connectivity channel
Web portal	Web	Follow-up / self-service
Other approved channels	—	System is explicitly extensible, not fixed to one intake point

6. End-to-End Scenario Walkthrough

A concrete example tying every piece together — useful to walk through as a team before diving into individual features.

Point in time	What happens
Day 0 — FIR registered	Victim files a complaint under the PoA Act through NHAA. A docket number is created in SAMBAL; this system enrolls the victim under the same docket number , and requests consent — audio-explained, in her preferred language — before any check-ins begin
Week 1-3 — Routine check-ins	Chatbot reaches out twice a week via the mobile app. Sentiment stays neutral-to-mild; Distress Score sits around 20-30/100 (Low risk)
Week 4 — Case event logged	A court hearing date is scheduled. Logged as a <code>case_timeline_events</code> entry — no distress-score effect yet on its own

Point in time	What happens
Week 4, two days later	Check-in text shows increased anxiety language; voice check-in shows elevated stress markers (Voice Stress Analytics). Score rises to 55 (Moderate)
Week 5 — Hearing postponed	The hearing is delayed. Correlated against the logged case event, Distress Score spikes to 78 (High) — well above her personal trend baseline
Alert triggered	System notifies the assigned counsellor and district welfare officer. Explainability panel shows exactly why: negative sentiment in the last 2 check-ins, elevated voice stress, and a hearing-delay event logged 2 days prior
Counsellor action	Counsellor reviews the case, contacts the victim directly, and logs a recommended intervention — counselling session + legal-aid follow-up. Marks the alert "confirmed, action taken"
Week 6 onward	Distress Score gradually declines back toward baseline after the counselling session and a new hearing date is confirmed

This single walkthrough exercises nearly every table in Section 9 and every model in Section 10 — worth using as the team's shared mental model of "what does this system actually do."

7. Mandatory Requirements (SIH's Expected Solution)

#	Requirement	Related Innovation Component(s)
1	Conduct periodic interactions via chatbot, IVRS, SMS, mobile app, web portal, or helpline follow-up	Multilingual Conversational AI
2	Analyse voice, text, behavioural responses, and engagement patterns	NLP, Sentiment Analysis, Emotion AI
3	Generate a Dynamic Distress Score with longitudinal trend analysis	Predictive Risk Modelling
4	Predict escalation before a crisis emerges — not just detect it after the fact	Predictive Risk Modelling
5	Trigger alerts to counsellors, district authorities, designated officials at risk thresholds	Real-Time Risk Alerts, Automated Case Prioritisation
6	Recommend interventions (counselling, medical, witness protection, relocation, financial aid, legal aid, rehabilitation)	—
7	Provide district / State / national-level dashboards	—
8	Ensure explainable AI, privacy protection, data security, legal/ethical compliance	Explainable AI

8. Feature Roadmap

Tier	Feature
Must-have	Multi-channel intake (chatbot + mobile/web minimum)
Must-have	NLP sentiment analysis on text/voice responses
Must-have	Dynamic Distress Score with trend tracking
Must-have	Threshold-based alerting to counsellors/officials
Must-have	Intervention recommendation mapped to risk level
Must-have	District/State/national dashboard (role-scoped)
Must-have	Basic explainability — surfaces why a score was flagged
Must-have	Consent/privacy flow
Must-have	2-3 Indian languages minimum
Additional	Voice Stress Analytics
Additional	Automated case-prioritisation queue
Additional	Per-victim history timeline with trend chart
Additional	Counsellor feedback loop (resolved/false-positive)
Additional	Role-based access control
Additional	Simulated NHAA/Integrated Portal API integration
Standout	Explainability shown visually (transcript spans / feature importance), not just claimed
Standout	Disengagement-as-signal — flags silence/withdrawal after high-risk contact, not only active distress
Standout	Case-timeline correlation — distress spikes overlaid against court dates/hearing delays
Standout	Counsellor AI copilot — auto-drafted case notes
Standout	Granular, revocable, audio-explained consent as a real screen

9. Feature Parity — Mobile App & Web App

Both platforms carry the same feature set — no functionality is exclusive to one app.

Feature	Mobile App	Web App
Multi-channel check-in (chatbot, voice)	Included	Included
Consent & privacy flow	Included	Included
Multilingual interface	Included	Included

Feature	Mobile App	Web App
Personal distress score & history view	Included	Included
Alert queue / notifications	Included	Included
Role-scoped dashboard	Included	Included
Intervention recommendation & tracking	Included	Included
Explainability view	Included	Included
Case-timeline correlation	Included	Included
Disengagement flagging	Included	Included
AI-assisted case notes	Included	Included

10. Competitive Landscape — What Already Exists

Layer	What exists	Gap
Tele-MANAS (14416)	Ministry of Health's national tele-mental-health helpline — 24/7 counselling, psychiatric consultation, AI chatbot triage, 20+ languages, 3.4M+ calls handled since 2022	General-purpose — doesn't know a caller is a PoA Act victim, doesn't track the same person over months tied to a legal case. A rising share of AI conversations are being handed off to humans because the AI underperforms on nuanced distress
NHAA / SAMBAL (14566)	MoSJE's own helpline + web portal for atrocity victims — tracks FIR registration, investigation/chargesheet status, grievance status lookup	Tracks the case, not the person — no mental-health score, no trend line, no alert if a victim is quietly deteriorating between hearings
Witness Protection Scheme, 2018	Supreme-Court-approved framework — categorized physical protection (police escort, identity change, relocation) for victims/witnesses	Only a physical-security guideline, not backed by law — weak implementation, uneven state rollout, limited funding; protection orders lapse after 3 months and need re-application. No continuous digital well-being layer at all
AI mental-health chatbots (India)	Wysa, InnerHour — culturally-aware conversational AI, CBT exercises, stigma-sensitive design	Consumer wellness apps — no case/legal-timeline awareness, no authority-escalation pipeline
Global AI crisis-detection tools	Crisis Text Line's ML model identifies ~85-90% of texters at severe/imminent risk from message text alone	Known false positives on ordinary language (e.g. flagging "not suicidal, just depressed" as high-risk), and bias against non-English/regional-language speakers — not tuned for Indian languages or the caste-violence context

Layer	What exists	Gap
Academic prototypes (2025)	Combine facial emotion recognition + speech emotion recognition + chat/scale-based assessment into one "holistic" monitoring pipeline	Research-stage, not deployed; not tied to any legal-case system

No system links case events to distress. A delayed chargesheet, an upcoming hearing, or the accused getting bail are exactly the moments distress is likely to spike — yet nothing above connects legal-case data with mental-health monitoring. **That is the one genuinely new piece this PS needs to deliver**, validated by every source reviewed here.

11. Integration & Value-Add Strategy

Extend existing government infrastructure rather than duplicating it:

Instead of building	Do this
A new parallel victim database	Plug into NHAA/SAMBAL and the Integrated Portal, reusing the same case ID so distress data links to real case milestones
A separate counsellor call centre	Reuse Tele-MANAS's existing counsellor network for escalation/hand-off
A text-only chat interface	Support regional languages and voice heavily — many victims are rural or semi-literate, so text-only excludes them
Compliance as an afterthought	Build in Explainable AI and DPDP Act 2023-compliant privacy from day one — this is caste-violence + mental-health data; a leak could endanger the victim
A generic distress score	Correlate hearing dates, bail status, and chargesheet delays with the distress score — the genuinely new piece no existing system currently does
A flat alert list	District/state dashboards letting SC/ST welfare officers prioritise the highest-risk victims first

12. Database Design (Scoped to Mandatory Features)

Normalized to 3NF, PostgreSQL, ACID-compliant.

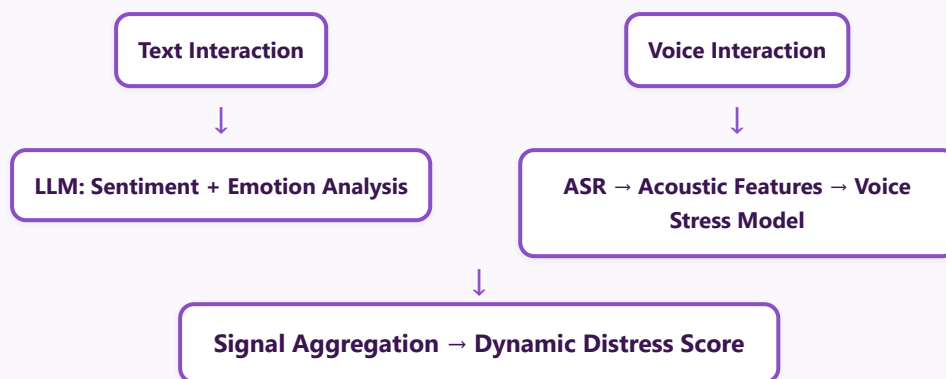
Table	Columns	Supports Requirement #
victims	victim_id, docket_number, case_type_id, jurisdiction_id, preferred_language, enrolled_at, status	1
channels	channel_id, channel_name	1

Table	Columns	Supports Requirement #
interactions	interaction_id, victim_id, channel_id, occurred_at, duration_seconds, transcript_ref	1
signal_types	signal_type_id, name	2
interaction_signals	signal_id, interaction_id, signal_type_id, value, confidence, model_version	2
risk_levels	risk_level_id, name, sort_order	3
distress_scores	score_id, victim_id, interaction_id, score_value, risk_level_id, model_version, computed_at	3, 4
alert_statuses	alert_status_id, name	5
alerts	alert_id, victim_id, distress_score_id, alert_status_id, triggered_at, resolved_at	5
alert_notifications	alert_id, official_id, notified_at	5
intervention_types	intervention_type_id, name	6
interventions	intervention_id, victim_id, alert_id, intervention_type_id, assigned_official_id, recommended_at, completed_at	6
jurisdictions	jurisdiction_id, name, level, parent_id	7
officials	official_id, full_name, role_id, jurisdiction_id, email, phone, password_hash	7
roles	role_id, role_name	7
victim_identity	victim_id, full_name, contact_number, address, id_ref_encrypted — kept separate from victims so the scoring/alert pipeline never touches raw PII	8
consent_records	consent_id, victim_id, channel_id, granted_at, revoked_at	8
audit_log	log_id, official_id, victim_id, action, entity_type, entity_id, occurred_at	8

Design notes: `victims.docket_number` is deliberately the same case ID used by NHAA/SAMBAL (Section 8) rather than a newly minted one, so distress data links directly to real case milestones instead of living in a parallel system. Predicted/forecasted scores reuse `distress_scores` with a distinct `model_version` rather than a separate table. Disengagement is computed via query (last interaction vs. now), not stored, since a derived fact kept in a table would need to be kept in sync manually — exactly what normalization avoids. `victim_identity` is deliberately separated from `victims` so the scoring/alert pipeline never touches raw PII.

13. AI/ML Model Architecture

The PS names 8 Innovation Components; as actual distinct models, this collapses to 4:



Component	Modality	Approach
Conversational chatbot + text sentiment/emotion	Text (generative + analysis)	One LLM, two prompts — same model drives dialogue and sentiment/emotion scoring via different system prompts
ASR + TTS + multilingual	Voice	One speech API pipeline (e.g. Google Speech-to-Text / Bhashini) — gets multilingual support without building it separately
Voice Stress Analytics	Voice (acoustic, not linguistic)	The one genuinely specialized model — cannot be substituted by an LLM. A pretrained speech-emotion-recognition model (e.g. wav2vec2-based) or hand-extracted acoustic features (pitch/jitter/pace) feeding a classifier. Reads <i>how</i> something is said, not <i>what</i>
Distress Score + escalation prediction	Structured / sequential	Weighted combination of sentiment + voice-stress + engagement signals, plus a trend check across recent check-ins

Not separate models: Explainable AI is a technique layered on the sentiment/scoring models (surfacing which signals drove a score), not its own model. Automated Case Prioritisation is sort/ranking logic on top of the distress score and case metadata — pure business logic, no model required.

External benchmark: Crisis Text Line's ML model identifies roughly 85-90% of texters at severe/imminent risk from text alone — a useful accuracy reference point for the text-analysis path, though it also carries known false-positive and language-bias limits (Section 7) that any Indian-language, caste-violence-context model needs to actively correct for, not inherit.

14. Training Data Strategy

Aspect	Approach
Real victim data	Not obtainable — ethically and legally out of reach; do not attempt to source it
Proxy datasets	GoEmotions (text emotion), DAIC-WOZ-style distress-interview data, RAVDESS/CREMA-D (speech emotion recognition)

Aspect	Approach
What proxy data validates	Pipeline mechanics work end to end — not a claim of domain-specific accuracy on this exact population
Synthetic personas	2-3 profiles with differing distress trajectories (stable / gradually escalating / sudden crisis), used in place of real case data
Production path	A real deployment would require an ethics-board-approved pilot with consenting NHAA users before any model claims domain accuracy

15. Legal, Privacy & Compliance

Requirement	How Addressed
DPDP Act 2023 — explicit, revocable consent	<code>consent_records</code> table, per-channel, with <code>revoked_at</code>
Purpose limitation & data minimization	<code>victim_identity</code> split from <code>victims</code> /scoring tables — scoring pipeline never touches raw PII
Breach-notification readiness	Append-only <code>audit_log</code> covering every access/action
Government data hosting	Production would need empanelled government cloud (NIC/MeitY "Meghraj" GI Cloud) rather than generic commercial cloud — an architecture constraint to design around, not solvable at the demo stage
Right to erasure	Cascading delete on <code>victim_identity</code> ; scoring/interaction history retained only in pseudonymous form for aggregate statistics

16. System Safety Design

Risk	Mitigation
False negative — a missed crisis	Conservative alert thresholds + mandatory human review before any action is taken
False positive — alert fatigue for counsellors	Confidence scoring per alert + counsellor feedback loop (resolved / false-positive) for ongoing calibration
Autonomous AI action on a person's safety	AI never acts alone — it recommends and prioritizes only; a human official always makes the final call. Matches the PS's own wording: "recommend," not "auto-dispatch"

17. Assumptions & Open Questions for Team Discussion

Worth debating as a team before locking in scope — none of these are answered by the PS text itself:

Assumption / Question	Why it matters
Victims have at least occasional phone access (own, family, or NGO-facilitated)	Core to whether mobile/IVRS/SMS are realistic entry points for this specific population
NHAA/SAMBAL's consent at initial complaint registration does not automatically cover this system's ongoing monitoring	Legal question — very likely needs its own explicit, separate consent step
Whether counsellors would be the existing Tele-MANAS network or a separate MoSJE-employed pool	Changes whether "reuse Tele-MANAS's counsellor network" (Section 8) is operationally real or just an architectural suggestion
Whether case-timeline events (hearing dates, bail status, chargesheet delays) are available as structured data from courts/police systems	If not, the case-timeline correlation feature — the single most-differentiating idea in this analysis — needs a manual-entry fallback
Team's current comfort with speech-emotion-recognition specifically	The one AI component that can't be shortcut with an LLM call (Section 10) — worth knowing who owns it early

18. Sources

- PIB — 14566: National Helpline Against Atrocities on SCs/STs (pib.gov.in)
- National Helpline Against Atrocities (NHAA / SAMBAL) portal (nhapoa.dosje.gov.in)
- Tele-MANAS official portal, Ministry of Health & Family Welfare (telemanas.mohfw.gov.in)
- India's Tele-MANAS: evolution, early outcomes — British Journal of Psychiatry / Cambridge Core
- Tele-MANAS callers turning to humans after AI falls short — Vision IAS / The Hindu, Apr 2026
- Speech Emotion Recognition in Mental Health: Systematic Review — JMIR Mental Health, 2025
- AI-Powered Holistic Mental Health Monitoring (FER + Chatbot + Voicebot) — ResearchGate, 2025
- Crisis Text Line — Detecting Crisis: An AI Solution (crisistextline.org)
- Crisis Text Line case study (bestpractice.ai)
- Scheduled Castes and Scheduled Tribes (Prevention of Atrocities) Act, 1989 (ncsk.nic.in)
- Witness Protection Scheme, 2018 — Challenges & Reforms — International Journal of Law Management & Humanities (ijlmh.com)
- Witness Protection Scheme: Why India needs it — ClearIAS